

数学分析(三).

数学分析(一): 一元实函数的微积分.

数学分析(三): 多元实函数的微积分.

数学分析(一)

$I \subset \mathbb{R}$: $f(x)$: $I \longrightarrow \mathbb{R}$
 $x \longmapsto y = f(x)$. 一元函数.

数学分析(三)

$x \in E \subset \mathbb{R}^n$ $f(x)$: $E \longrightarrow \mathbb{R}$ 多元函数.
 $x = (x_1, x_2, \dots, x_n)$ $x = (x_1, x_2, \dots, x_n) \longmapsto u = f(x)$

$\mathbb{R}$ 与 $\mathbb{R}^n$ 之差别

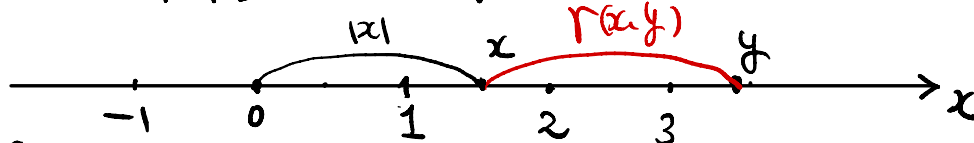
$\mathbb{R}$: 实数集. $\mathbb{R} = \{x \mid -\infty < x < \infty \text{ 实数} \}$

四则运算: $+$, $-$, $\times$, $\div$. $x, y \in \mathbb{R}$

序: 比较大小. $\Rightarrow x \cdot y \in \mathbb{R}, x \div y \in \mathbb{R}$
 $y \neq 0$

Descartes: $\mathbb{R}$: 实数集 $x, y \in \mathbb{R}$
 (笛卡尔) $\updownarrow$ 对应 $x < y$

有向直线 —— 数轴



$|x|$: 绝对值.

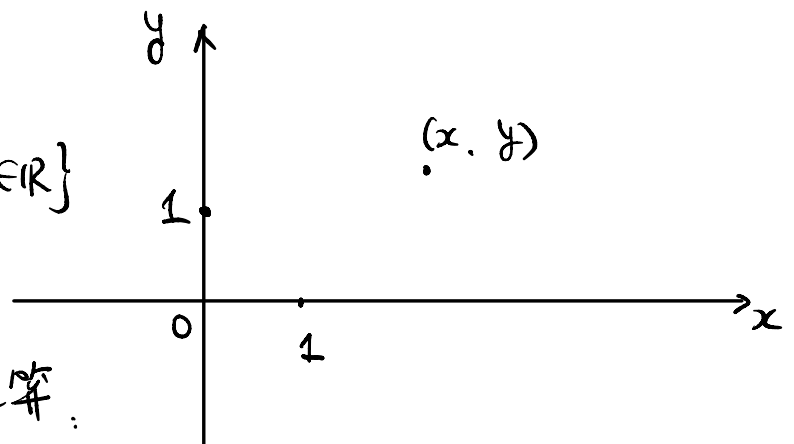
x, y 之间之距离: $r(x, y) = |x - y|$

$$\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}, \quad \left(\text{集合乘积: } A \times B = \{ (a, b) \mid a \in A, b \in B \} \right. \\ \left. = \{ (x, y) \mid x \in \mathbb{R}, y \in \mathbb{R} \} \neq B \times A \right)$$

$$\mathbb{R}^n = \overbrace{\mathbb{R} \times \mathbb{R} \times \cdots \times \mathbb{R}}^{n \text{ 重}}$$

$$= \{ x = (x_1, x_2, \dots, x_n) \mid x_i \in \mathbb{R} \}$$

—— $\mathbb{R}$ 上的 n 重直积



$\mathbb{R}^n$ 上的加法、数乘运算:

$$x = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n,$$

$$x_i, y_i \in \mathbb{R}, \quad \lambda \in \mathbb{R}$$

$$y = (y_1, y_2, \dots, y_n) \in \mathbb{R}^n.$$

$$x + y = (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n) \in \mathbb{R}^n. \quad \text{加法运算封闭}$$

$$\lambda x = (\lambda x_1, \lambda x_2, \dots, \lambda x_n) \in \mathbb{R}^n. \quad \text{数乘运算封闭}$$

$$x + 0 = x$$

$$\lambda = -1, \quad (-1) \cdot x = -x$$

$$x + (-x) = 0$$

$\mathbb{R}^n$: $\mathbb{R}$ 上的 n 维线性空间

$$x = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n, \quad \text{点}, \quad n \text{ 维向量}, \quad x_i: \text{第 } i \text{ 个分量}$$

$$\|x\| = \sqrt{\sum_{i=1}^n x_i^2}: \quad x \text{ 的 "大小" (norm)}$$

$$\text{距离 (度量 metric): } \forall x, y \in \mathbb{R}^n. \quad x = (x_1, x_2, \dots, x_n) \\ -y = (\underbrace{y_1}_{-}, \underbrace{y_2}_{-}, \dots, \underbrace{y_n}_{-})$$

$$r(x, y) = \|x - y\|$$

$$x - y = x + (-y)$$

$$= \|(x_1 - y_1, x_2 - y_2, \dots, x_n - y_n)\|$$

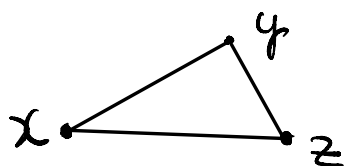
$$= \sqrt{\sum_{i=1}^n (x_i - y_i)^2} = \sqrt{\sum_{i=1}^n (y_i - x_i)^2}$$

性质: ① 正定性: $r(x, y) \geq 0$; $r(x, y) = 0 \Leftrightarrow x_i = y_i, 1 \leq i \leq n$.
 $\Leftrightarrow x = y$.

② 对称性: $\forall x, y \in \mathbb{R}^n \quad r(x, y) = r(y, x)$

③ 三角不等式: $\forall x, y, z \in \mathbb{R}^n$:

$$r(x, y) \leq r(x, z) + r(z, y)$$



两边之和 $\geq$ 第三边.

④ $\|\lambda x\| = |\lambda| \|x\|$. ———— 正齐性;

$$\begin{aligned} r(\lambda x, \lambda y) &= \|\lambda x - \lambda y\| = \|\lambda(x - y)\| = |\lambda| \|x - y\| \\ &= |\lambda| r(x, y) \end{aligned}$$

⑤: $r(x - x', y - x') = r(x, y)$ ———— 平移不变性

$$= \|(x - x') - (y - x')\| = \|x - y\| = r(x, y)$$

$\mathbb{R}^n$: 赋予代数结构与拓扑结构的 $\mathbb{R}^n$ 称为欧几里得空间.
 欧氏空间.

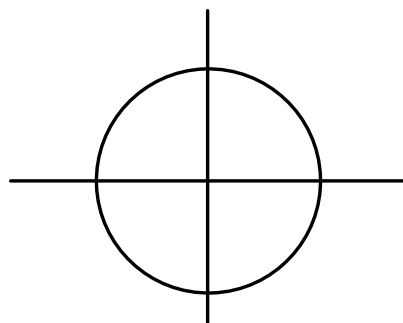
$$\|x\|_1 = \sum_{i=1}^n |x_i|; \quad \|x\|_\infty = \max_{1 \leq i \leq n} |x_i|;$$

$(\mathbb{R}^n, \|\cdot\|_1)$ $(\mathbb{R}^n, \|\cdot\|_\infty)$: ———— 不是欧氏空间

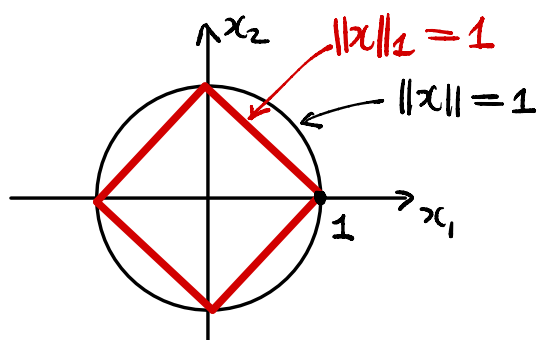
$(\mathbb{R}^n, \|\cdot\|)$: 欧氏空间.

$$x \in \partial B(0, 1): \Leftrightarrow \|x\| = 1.$$

$$x_1^2 + x_2^2 + \dots + x_n^2 = 1.$$



$$\|x\|_1 = 1, \quad \sum_{i=1}^n |x_i| = 1, \quad x_1 + x_2 + \dots + x_n = 1, \quad x_i > 0;$$



“内积” $x, y \in \mathbb{R}^n$,

$$x = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n,$$

$$y = (y_1, y_2, \dots, y_n) \in \mathbb{R}^n.$$

$$\langle x, y \rangle = \sum_{i=1}^n x_i y_i. \quad \text{—— 称为 } x \text{ 与 } y \text{ 的内积.}$$

$$\textcircled{1} \quad \langle x, x \rangle = \sum_{i=1}^n x_i \cdot x_i = \left(\sum_{i=1}^n x_i^2 \right) = \|x\|^2 \geq 0;$$

$$\langle x, x \rangle = 0 \iff x = 0; \quad \text{—— 正定性}$$

$$\textcircled{2} \quad \langle x, y \rangle = \langle y, x \rangle, \quad \text{—— 对称性}$$

③. 关于第三个量 z 是线性的:

$$\langle x+y, z \rangle = \langle x, z \rangle + \langle y, z \rangle$$

$$\langle \lambda x, z \rangle = \lambda \langle x, z \rangle,$$

} Easy to check!

④ Cauchy-Schwarz 不等式:

$$|\langle x, y \rangle| \leq \|x\| \cdot \|y\|.$$

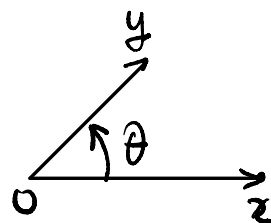
$$|\langle x, y \rangle| = \left| \sum_{i=1}^n x_i y_i \right| \leq \sqrt{\sum_{i=1}^n x_i^2} \cdot \sqrt{\sum_{i=1}^n y_i^2}.$$

$$\left| \frac{\langle x, y \rangle}{\|x\| \cdot \|y\|} \right| \leq 1 \Rightarrow \theta \in [0, \pi], \quad \cos \theta = \frac{\langle x, y \rangle}{\|x\| \cdot \|y\|}.$$

θ : 称为 x 与 y 之间的夹角.

$$\langle x, y \rangle = 0 \iff \cos \theta = 0$$

$$\iff \theta = \frac{\pi}{2} \iff x \perp y$$



$\mathbb{R}^n$ 中 $\leq$ 是 $\frac{1}{2}$.

1. 令 $\mathcal{A}$ 是 $\mathbb{R}^n$.

1. $M_0(x_1^{(0)}, x_2^{(0)}, \dots, x_n^{(0)}) \in \mathbb{R}^n$

$$\delta > 0 \quad \sqrt{\sum_{i=1}^n (x_i - x_i^{(0)})^2} < \delta$$

$$x = (x_1, x_2, \dots, x_n)$$

$$O(M_0, \delta) = \{x = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n \mid r(x, M_0) < \delta\}$$

$$= \{x = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n \mid \sum_{i=1}^n (x_i - x_i^{(0)})^2 < \delta^2\}$$

—— $\Rightarrow M_0$ 为中心 (球心) δ 为半径的 $\mathbb{R}^n$ 中球.

或: M_0 是 δ 邻域

或: $\Rightarrow M_0$ 为球 δ 为半径的 球.

ball centered at M_0 of radius δ

$$O(M_0, \delta) = B(M_0, \delta)$$

不需要知道 δ 之大小: 简记为 $O(M_0)$, 或 $B(M_0)$

$$\overset{\circ}{O}(M_0, \delta) = O(M_0, \delta) \setminus \{M_0\}: \text{空心邻域, 空心邻域}.$$

2. $\frac{1}{2}$ $\mathbb{R}^n$:

$$x^{(k)} \in \mathbb{R}^n: k=1, 2, \dots$$

$$x^{(k)} = (x_1^{(k)}, x_2^{(k)}, \dots, x_n^{(k)}), k=1, 2, \dots$$

$$\{x^{(k)} \mid k \geq 1\} \subset \mathbb{R}^n: \text{—— } \frac{1}{2} \mathbb{R}^n.$$

$$M_0(x_1^{(0)}, x_2^{(0)}, \dots, x_n^{(0)}) \in \mathbb{R}^n.$$

若 $r(x^{(k)}, M_0) \xrightarrow{k \rightarrow \infty} 0$; 则称点列 $\{x^{(k)} | k \geq 1\}$ ($= \{x^{(k)}\}$)

收敛于 M_0 . 或 $\{x^{(k)}\}$ 之极限点列. 记 $\lim_{k \rightarrow \infty} x^{(k)} = M_0$.

$$\lim_{k \rightarrow \infty} x^{(k)} = M_0 \Leftrightarrow \forall \varepsilon > 0, \exists N > 0, \rightarrow \forall k > N, \text{ 有}$$

$$r(x^{(k)}, M_0) = \sqrt{\sum_{i=1}^n (x_i^{(k)} - x_i^{(0)})^2} < \varepsilon$$

$$\Leftrightarrow \forall \varepsilon > 0, \exists N > 0, \rightarrow \forall k > N, \quad \sum_{i=1}^n (x_i^{(k)} - x_i^{(0)})^2 < \varepsilon^2.$$

$$\left(\Rightarrow \begin{array}{c} \forall (1 \leq i \leq n, |x_i^{(k)} - x_i^{(0)}| < \varepsilon \\ \text{注意表述} \end{array} \right)$$

$$\Leftrightarrow \lim_{k \rightarrow \infty} x_i^{(k)} = x_i^{(0)} \quad (1 \leq i \leq n).$$

性质: $\lim_{k \rightarrow \infty} x^{(k)} = M_0 \Leftrightarrow \forall (1 \leq i \leq n), \quad \lim_{k \rightarrow \infty} x_i^{(k)} = x_i^{(0)}.$

$\Updownarrow$
按分量收敛.

性质: 收敛点列之极限是唯一的. (Ex.)

$\mathbb{R}$ 之完备性: Cauchy 数列都有极限.

七大定理.

$\mathbb{R}^n$ 之完备性: ?